POLICY BRIEF · TEACHER LABOR MARKETS · JULY 2026

Who leaves teaching? Two decades of teacher attrition in the United States, 2005–2025

Evidence from 174,872 school teachers followed for twelve months each in linked Current Population Survey microdata

EXECUTIVE SUMMARY

- Because **one in six 12-month leavers (15.7%) is back teaching within three months**, we define attrition strictly: a *persistent leaver* is out of teaching twelve months later *and* in every re-interview thereafter. By this measure, **15.1% of degree-holding school teachers persistently leave the profession each year**: 9.9 percentage points (pp) to another occupation, 4.5 pp out of the labor force, 0.6 pp to unemployment. The conventional 12-month definition gives 17.6%.
- Persistent attrition has **risen by about four points in two decades** — from 13% in 2005–2010 to 17% in 2022–2024, with a COVID spike (18.4% in 2019→2020) and a new high of 18.0% in 2024.
- **Working part-time is by far the strongest predictor** of leaving ($+12.2$ pp). Exit risk is U-shaped in age (lowest at 40) and higher for Black ($+5.9$ pp), non-citizen ($+7.6$ pp) and preschool/kindergarten ($+4.2$ pp) teachers. Among degree holders, a graduate degree adds only modest extra protection (-1.1 pp). All effects are nearly identical under the conventional definition — the predictors are robust.
- The **typical persistent leaver** is a woman in her late career, working part-time, without a graduate credential — the top predicted-risk decile faces a 35% exit probability, 2.4 times the average teacher.

15.1 %
of teachers leave the profession for good each year

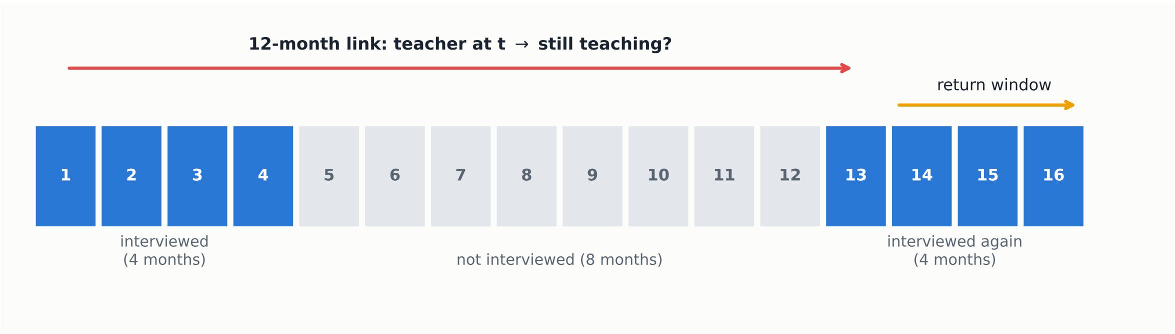
$+$12.2 pp
extra exit risk from part-time work

1 in 6
12-month leavers is back teaching within 3 months

1. The population we observe — who can be followed, and for how long

We use official **Current Population Survey (CPS) basic monthly microdata**: 251 monthly files covering January 2005 to December 2025, from the U.S. Census Bureau (2024–2025) and the NBER mirror (2005–2023); October 2025 was never released. The CPS interviews each household on a **4–8–4 rotation** (Figure 1): four consecutive months in sample, eight months out, four months back in — so any person is observable for at most **16 months**, once. Within that window, each adult interviewed in year t is re-interviewed in the same calendar month of $t{+}1$ and can be linked by household and person identifiers, with matches validated on sex, race and age progression (Madrian–Lefgren criterion). No one can be followed across multiple years: each base year is a *fresh cohort*, which is why every rate in this brief is computed wave by wave.

Figure 1: The CPS follows each person for at most 16 months. Interview calendar of one rotation group; numbers are months since first interview.



The 12-month link compares months m and $m{+}12$ ($m \leq 4$); the return window re-observes leavers in months $m{+}13$ to 16.

Table 1 shows the full population funnel, pooling all 251 months.

Table 1: From the CPS universe to the analytic sample.

	Person-months	% of previous
Adult interviews (person-months), 2005–2025	23,752,044	–
employed	14,450,470	61%
school teachers (occ. 2300–2330)	525,651	4%
with bachelor's degree or higher	464,144	88%
in a linkable rotation (MIS 1–4)	229,930	50%
Linked to their interview 12 months later	174,872	–
with re-interviews after $t{+}12$ (MIS 1–3) $\rightarrow$ **main sample**	129,897	74%

Teachers are identified as employed persons whose primary-job occupation is preschool/kindergarten, elementary/middle, secondary, or special-education teacher (Census codes 2300–2330, identical across the 2002, 2010 and 2020 classifications), holding a bachelor's degree or higher. Only rotations in months-in-sample 1–4 can be linked forward; 67% of linkable teachers are found and validated 12 months later.

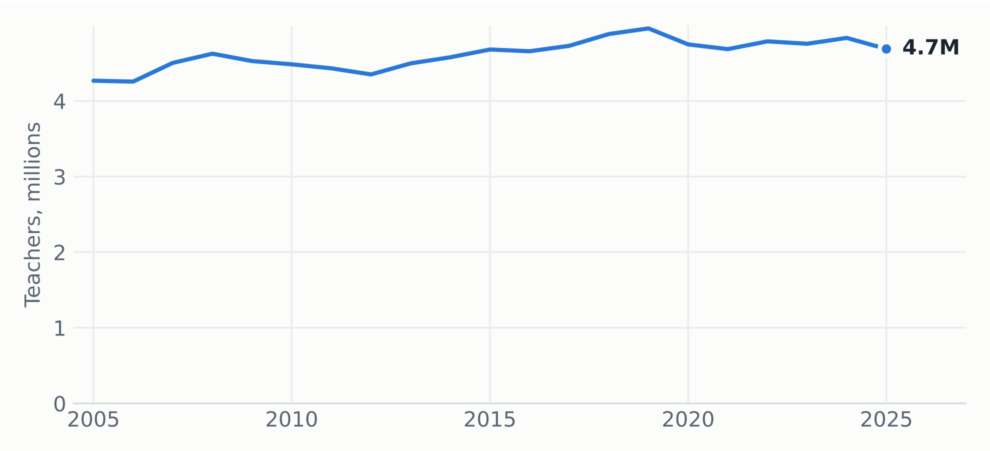
Defining a leaver. The conventional measure (*12-month leaver*, as in the NCES Teacher Follow-up Survey) counts anyone not employed as a school teacher twelve months after baseline. But temporary exits — parental leave, short unemployment spells, sabbaticals — inflate it: re-observing leavers in their up-to-three remaining monthly interviews, 15.7% are already teaching again. Our main measure is therefore the *persistent leaver*: out of teaching at $t{+}12$ *and* in every subsequent re-interview, computed on the 129,897 teachers whose rotation allows at least one re-interview. Exits longer than a year that end in a return cannot be detected — persistent attrition remains a (much tighter) upper bound on permanent exit. Rates use CPS final person weights.

Caveat: occupation in the returning rotation is re-coded independently, which inflates measured occupation switching; levels should be read as upper bounds, while *comparisons across groups and over time* — the object of this brief — are far less affected.

2. The teacher workforce: stock, composition, flows

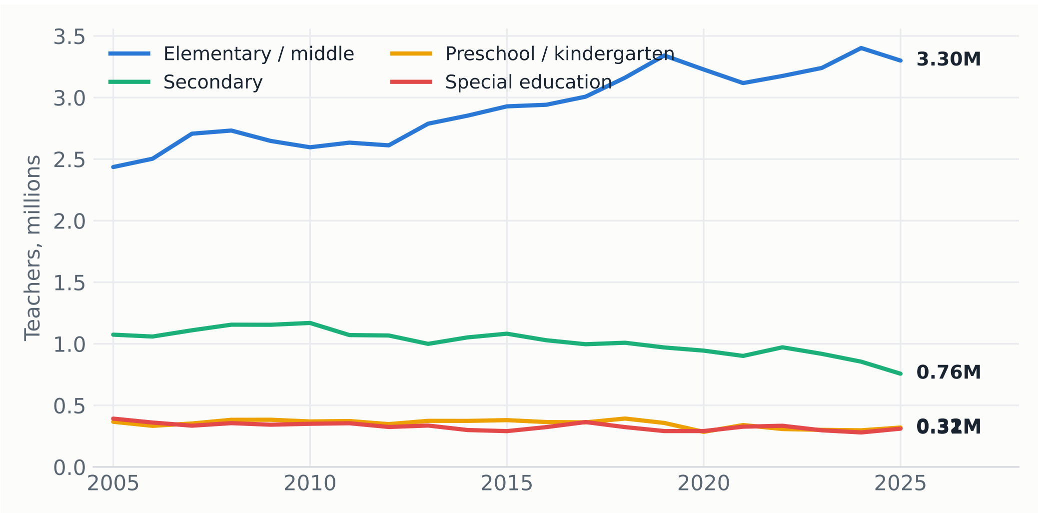
Weighted to population, the CPS counts a stable stock of degree-holding school teachers (Figure 2), dominated by elementary/middle school teaching (Figure 3).

Figure 2: The stock of school teachers. Estimated number of employed school teachers with a bachelor's degree or higher, annual average of monthly estimates.



Source: CPS basic monthly files, person weights. Teachers = employed, primary-job occupation codes 2300–2330 (preschool/kindergarten, elementary/middle, secondary, special education), bachelor's degree or higher.

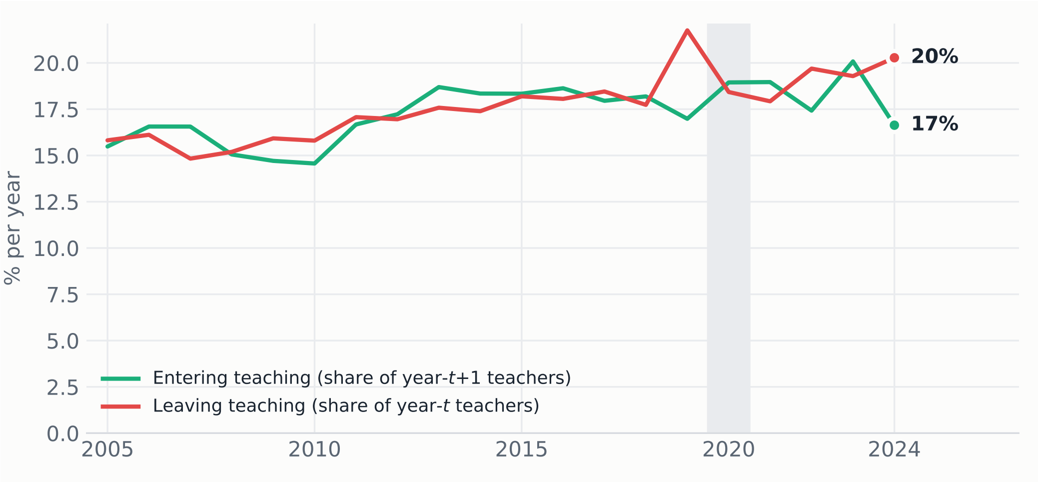
Figure 3: Teachers by teaching level. Same identification, by occupation code.



Source: CPS basic monthly files, person weights.

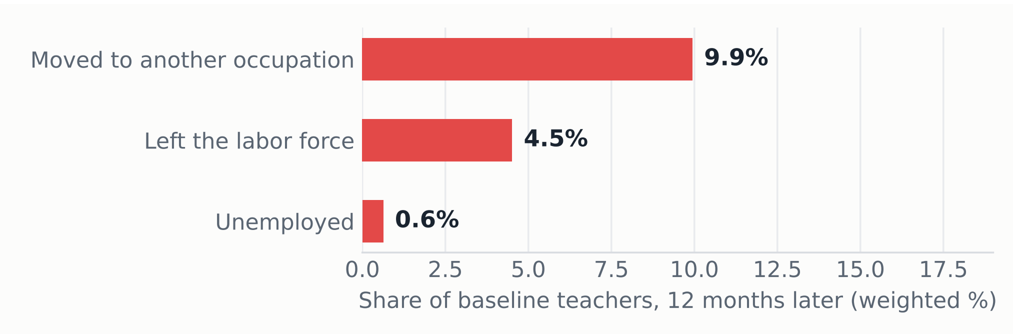
The linked pairs also reveal both sides of the turnover ledger (Figure 4): the profession simultaneously loses and recruits 15–20% of its stock every year — a high-churn occupation in which exits have been trending above entries in recent years.

Figure 4: Teaching loses and recruits 15–20% of its stock every year. Gross annual flows into and out of school teaching, linked pairs (12-month definitions).



Source: linked CPS pairs. Entering = school teacher (BA+) at $t{+}12$ who was not a school teacher at t; leaving = the reverse. Linked samples under-count movers, so both flows are conservative.

Figure 5: Most persistent leavers move to other jobs rather than out of work. Destination of persistent leavers, share of baseline teachers (weighted), 2005–2025.



Source: authors' calculations, linked CPS basic monthly files 2005–2025.

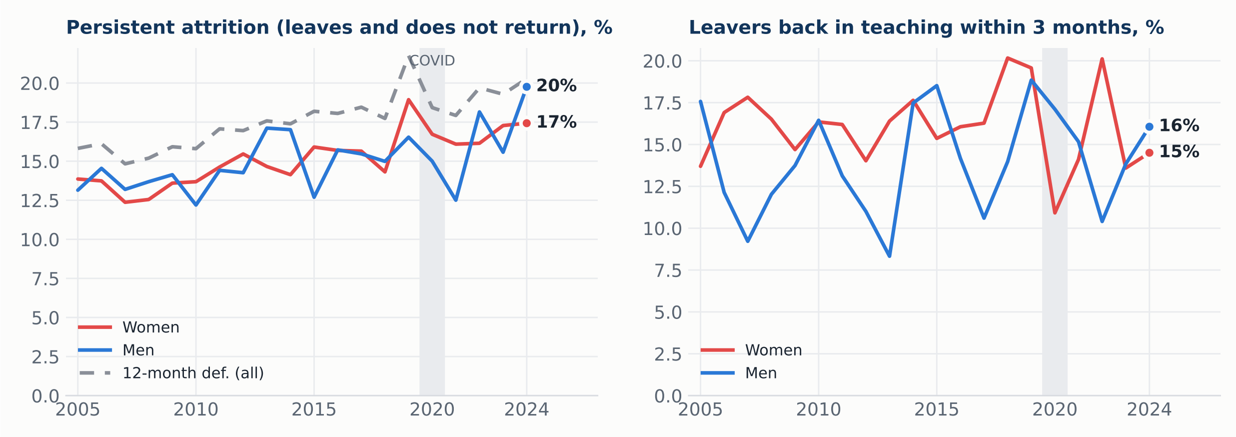
3. Attrition is rising — and part of it is churn

One in six 12-month leavers is teaching again within three months (women 16.1%, men 14.0%; Figure 6, right) — temporary exits that the persistent definition strips out. Net of this churn, **persistent attrition still climbed from 13% in 2005–2010 to 17% in 2022–2024** (Figure 6, left), peaking in the pandemic year (18.4% in 2019→2020) and reaching a new high of 18.0% in 2024. Gender gaps are small and unstable; in the most recent year men (19.8%) exceeded women (17.4%).

4. Who leaves: raw differences

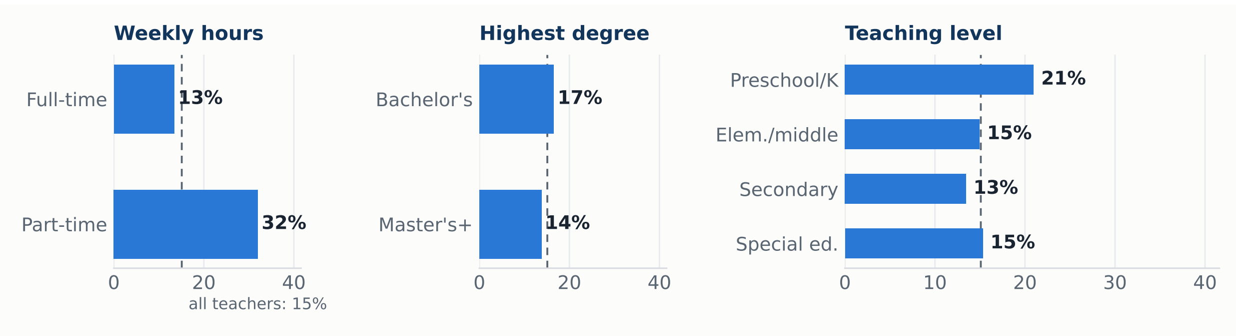
Attrition concentrates in three margins: **hours**, **teaching level** and, more weakly once degree holders only are considered, **credentials.** Part-time teachers leave at more than twice the rate of full-time teachers; preschool/kindergarten is the most exposed level.

Figure 6: Persistent attrition has risen four points in two decades; one in six 12-month leavers returns within three months. Weighted rates by base year and gender.



Source: authors' calculations, linked CPS 2005–2025. Return window truncated by the CPS design at three monthly interviews after $t+12$ (leavers re-interviewed at least once: 21,873 of 30,244).

Figure 7: Attrition more than doubles without full-time hours. Weighted persistent attrition rate by baseline characteristics, 2005–2025.



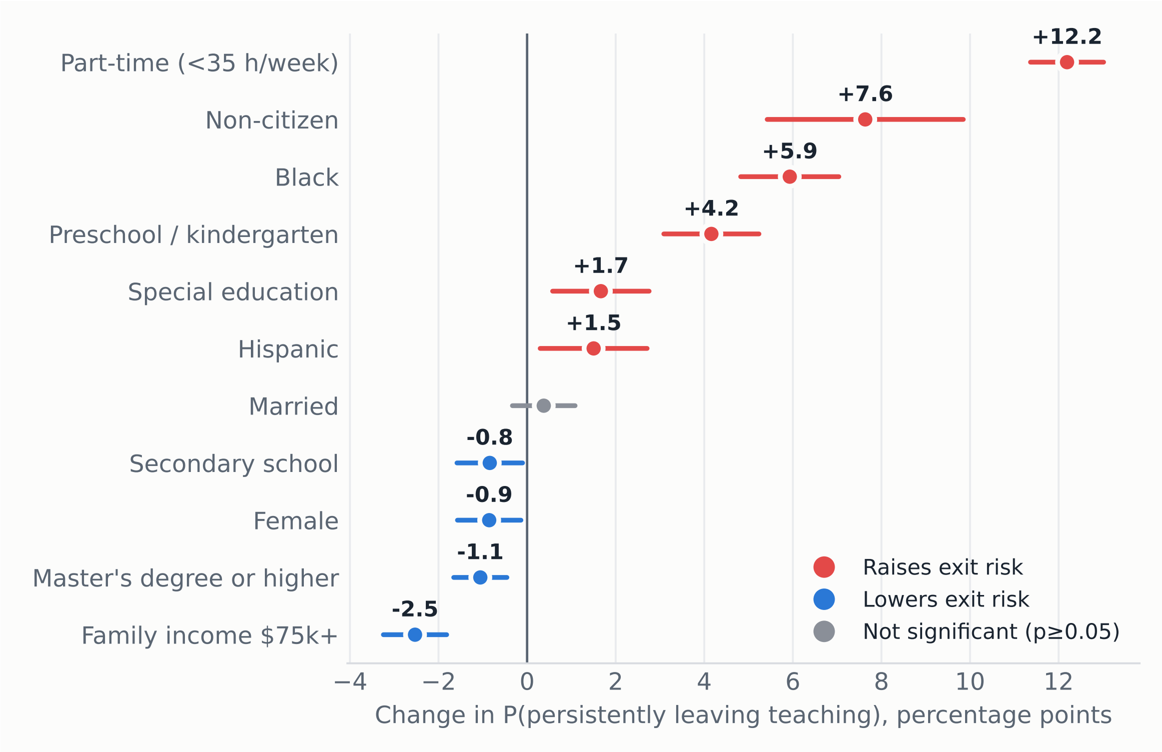
Source: authors' calculations, linked CPS 2005–2025. Dashed line: all-teacher average (15.1%).

5. A probit model of leaving teaching

We estimate a probit for the probability of *persistently* leaving, with covariates measured at baseline, base-year fixed effects, and standard errors clustered by household (Table 2); Figure 8 plots the average marginal effects. Three results stand out. First, **part-time status raises exit probability by 12.2 pp**, the largest effect in the model by an order of magnitude — attachment to full-time hours, not demographics, is the dominant margin. Second, **minority and non-citizen teachers are significantly more likely to leave** (Black $+5.9$ pp, non-citizen $+7.6$ pp, Hispanic $+1.5$ pp), a retention gap with direct implications for workforce diversity. Third, among degree holders **a graduate degree adds only -1.1 pp of protection**, while higher family income (-2.5 pp), teaching secondary school (-0.8 pp) and being a woman (-0.9 pp) also retain modestly. Every effect is nearly identical under the conventional 12-month definition (largest difference: part-time, $+14.5$ vs $+12.2$ pp) — the predictors are not an artifact of how leaving is measured.

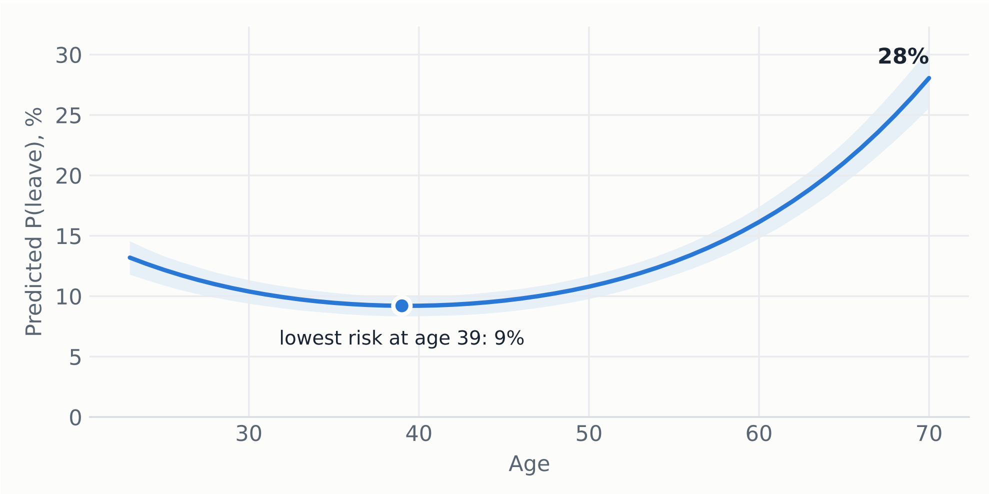
Age enters with a robust U-shape (Figure 9): predicted risk falls to a minimum of 9% around age 39 and accelerates after 55 as retirement margins open, reaching 28% at 70 — the familiar U-shaped attrition profile of the teacher-turnover literature.

Figure 8: Hours dominate demographics. Average marginal effects on the probability of leaving teaching within 12 months, with 95% confidence intervals.



Source: probit of Table 2. Age terms shown in Figure 9.

Figure 9: Exit risk is U-shaped in age, lowest near 40. Predicted probability of leaving by age, other covariates at sample means; shaded band: 95% CI.



Source: probit of Table 2, simulation-based confidence band.

6. The profile of the typical leaver

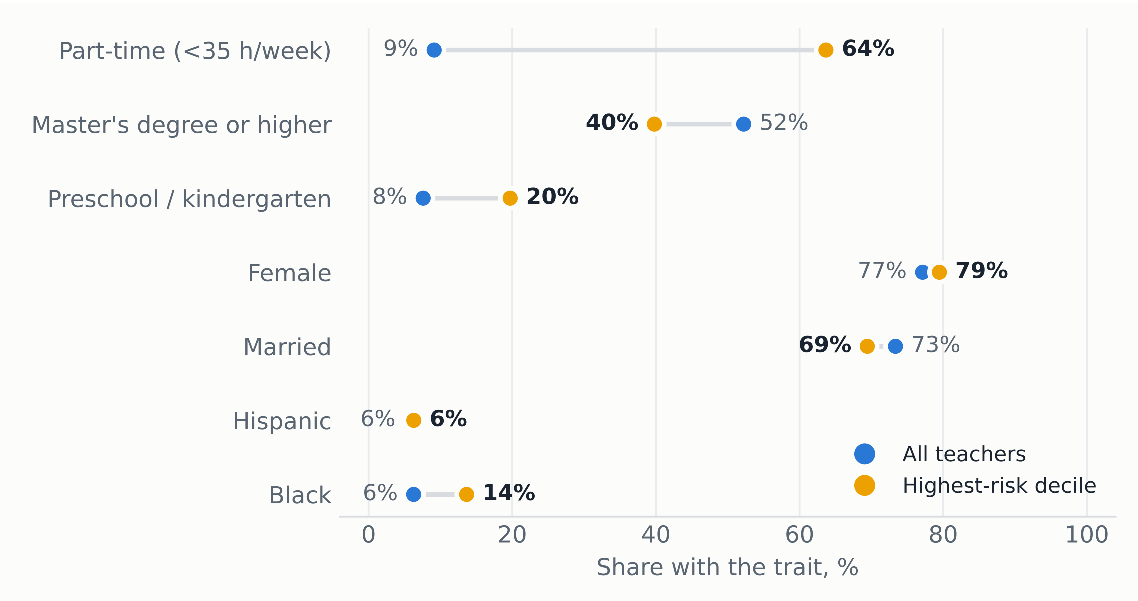
Ranking teachers by predicted risk, the top decile — with a mean persistent exit probability of 35%, against 15% for the average teacher — is overwhelmingly female (80%), in the late-career age range (mean 56), seven times more likely to work part-time (64% vs. 9%), less likely to hold a graduate degree (40% vs. 52%), and over twice as concentrated in preschool/kindergarten (20% vs. 8%), with Black teachers over-represented (14% vs. 6%).

Table 2: Probit estimates: probability of persistently leaving teaching, 2005–2025.

	Probit coef.	(SE)	AME (pp)
Age (years)	-0.062***	(0.004)	-1.36***
Age^2/100	0.079***	(0.004)	+1.73***
Female	-0.039**	(0.017)	-0.86**
Married	0.017	(0.017)	+0.38
Black	0.273***	(0.026)	+5.93***
Hispanic	0.069**	(0.028)	+1.50**
Non-citizen	0.351***	(0.052)	+7.63***
Master's degree or higher	-0.049***	(0.014)	-1.06***
Part-time (<35 h/week)	0.561***	(0.020)	+12.19***
Hours not reported	0.334***	(0.022)	+7.26***
Family income $75k+	-0.116***	(0.017)	-2.53***
Preschool / kindergarten	0.191***	(0.025)	+4.16***
Secondary school	-0.039**	(0.017)	-0.84**
Special education	0.077***	(0.026)	+1.67***
Constant	-0.039	(0.090)	
Base-year fixed effects		Yes	
Observations		129,897	
Pseudo R^2		0.055	

Outcome: persistent leaver. Cluster-robust standard errors (by household) in parentheses. AME: average marginal effect in percentage points. Sample: school teachers with a bachelor's degree or higher and a potential post-$t+12$ re-interview (baseline MIS 1–3). Reference categories: full-time, bachelor's degree, elementary/middle school. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 10: The highest-risk teacher: part-time, less credentialed, early-years. Characteristics of the top predicted-risk decile vs. all teachers.



Source: predicted probabilities from the probit of Table 2.

7. Policy implications

The results locate teacher attrition in **weak attachment** — part-time hours, the early-childhood segment, both career extremes — rather than in broad demographics, and show a **structurally rising trend** since the early 2010s. Four levers follow. (i) **Full-time conversion** of involuntary part-time positions attacks the single largest risk factor. (ii) **Retention packages for preschool**

and kindergarten teachers, the most exposed level. (iii) The significant **Black, Hispanic and non-citizen retention gaps** warrant targeted monitoring, since they compound into the diversity of the future workforce. (iv) With **one in six leavers back within three months**, cheap re-entry pathways — fast-track rehiring, credential portability across districts, return bonuses — can recover part of the outflow at low cost.

Methods in one line. Twenty linked CPS 4–8–4 rotation panels, 2005→2025 · 174,872 teachers (129,897 with follow-up) · persistent-leaver outcome · probit, year FE, household-clustered SE · fully reproducible: `src/02_build_cps_panel.py` → `src/05_evolution.py` → `src/03_model_attrition.py`.